

Multiple Choice (10 marks)

1. Find the Laplace transform of $f(t) = t$, for $t > 0$.
 - (a) $L\{t\} = (2 / s^3)$
 - (b) $L\{t\} = (1 / s^3)$
 - (c) $L\{t\} = s^2$
 - (d) $L\{t\} = (1 / (2 s^2))$
 - (e) $L\{t\} = (1 / s^2)$
2. A combinational PLD with a fixed AND array and a programmable OR array is called a
 - (a) ROM
 - (b) FPGA
 - (c) PLA
 - (d) PLD
 - (e) EEPROM
3. Find the energy stored in a nonlinear capacitor with characteristics given by $g(t) = 0.5 \nu^2(t)$ and $\nu(t) = 1 + 0.5 \sin t$ from $t = 0$ to $t = 1$ sec.
 4. 0.785 J
 5. 0.211 J
 6. 0.125 J
 7. 0.623 J
 8. 0.956 J

An electromagnetic wave of 40 KHz is radiated by an antenna . Find the time taken by this wave to reach a point 60 km away.

- (a) 0.3 msec
- (b) 0.8 msec
- (c) 0.6 msec
- (d) 0.25 msec
- (e) 0.1 msec

Compute the impact stress in a steel rod which is hit by a block weighing 45 lbs. The speed of the block is 40 in./sec. The rod is 18 in. long and has a diameter of $1\frac{1}{4}$ in.

- (a) 8,762.3 psi
- (b) 4,321.0 psi
- (c) 25,678.9 psi
- (d) 6.25 lbs
- (e) 30,000 psi

True / False (10 marks)

Answer True or False

- 6. True or False: The roots of the polynomial $s^4 + 3s^3 + 2s^2 + 2s + 1 = 0$ include -0.6008, -2.4486, and $0.0247 + j 0.824$, and $0.0247 + j 0.824$.
- 7. True or False: In an SR latch built from NOR gates, the condition S=1 and R=1 is not allowed as it leads to an undefined state.
- 8. True or False: The probability of obtaining either a 2 or a 5 in a single throw of a die is $\frac{1}{3}$.
- 9. True or False: The short circuit level at the busbar for four identical alternators rated 20 MVA each is 600 MVA.
- 10. True or False: The inverse Laplace transform of $F(s) = \frac{(2s^2 + 13s + 17)}{\{(s + 1)(s + 3)\}}$ yields $f(t) = 2(t) + (3e^{-t} + 2e^{-3t})u(t)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the calculation of the ratio of normalized power between the second and first harmonics of a periodic signal, using the provided formula for total normalized power.
- 12. Calculate the energy stored in an inductor from $t = 0$ to $t = 1$, given its i -characteristic $i(t) = \frac{1}{3}i^3(t)$ and the current $i(t) = 2 \sin(t)$. Discuss your methodology and results.

End of Paper